

Infinite sums and random points

A guided tour of the BeamerJS class



YOUR LOGO

First Last

Position

Department of Mathematics
University

Department seminar
October 6, 2026

Outline

1. The harmonic series
2. The Basel problem
 - Convergence
 - Euler's answer
3. Random points
4. Toolbox

1

The harmonic series

Partial sums

The basic vocabulary

Definition (Series). Let $(a_n)_{n \geq 1}$ be a real sequence. The **series** $\sum a_n$ **converges** if the sequence of **partial sums**

$$S_N = \sum_{n=1}^N a_n$$

has a finite limit as $N \rightarrow \infty$.

 We write $H_N = \sum_{n=1}^N \frac{1}{n}$ for the N th harmonic number.

A divergent series

Theorem (Oresme, around 1350). The harmonic series $\sum \frac{1}{n}$ diverges.

A divergent series

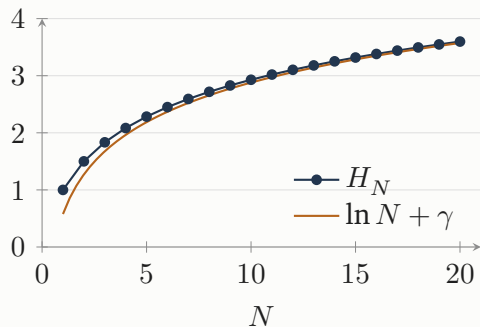
Theorem (Oresme, around 1350). The harmonic series $\sum \frac{1}{n}$ diverges.

Proof. Group the terms in blocks of length 2^k :

$$H_{2^m} = 1 + \frac{1}{2} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{\geq 1/2} + \underbrace{\frac{1}{5} + \dots + \frac{1}{8}}_{\geq 1/2} + \dots \geq 1 + \frac{m}{2}.$$

The partial sums are therefore **unbounded**. □

Logarithmic growth



Proposition.

$$H_N = \ln N + \gamma + o(1),$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant.

It diverges, but very slowly: $H_N > 10$ takes more than 12 000 terms.

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The Basel problem

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The squares converge

Lemma. For every $n \geq 2$, $\frac{1}{n^2} < \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$.

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Corollary. $\sum_{n=1}^{\infty} \frac{1}{n^2} < 1 + \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = 2$.

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Remark. The telescoping sum only gives a *bound*. The exact value is another story.

An open question in 1650

Conjecture (Mengoli, 1650). The sum $\sum_{n \geq 1} 1/n^2$ has a closed form.

Jacob Bernoulli, in Basel, could not crack it. The problem has carried the name of his city ever since.

“*Read Euler, read Euler, he is the master of us all.*

— attributed to Pierre-Simon Laplace

The exact value

Theorem (Euler, 1734 [2]).

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Beamer block

Standard blocks follow the palette.

Alert block

Euler's first argument was not fully rigorous.

Example block

Modern proofs can be found in [1].

A numerical check

N	$S_N = \sum_{n \leq N} 1/n^2$	$\pi^2/6 - S_N$
1	1.000 000	0.644 934
10	1.549 768	0.095 166
100	1.634 984	0.009 950
1000	1.643 935	0.001 000

- The error is about $1/N$.

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- Each correct digit costs **ten times more** terms.
- Euler accelerated the convergence and found $\pi^2/6 \approx 1.644\,934$ by hand.

Euler's idea

💡 Treat $\frac{\sin x}{x}$ as a polynomial whose roots $\pm\pi, \pm2\pi, \dots$ are known.

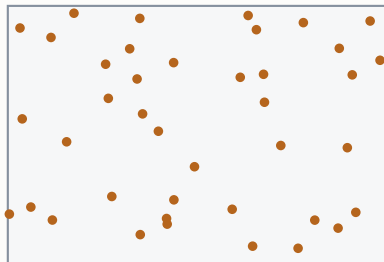
1. Factor: $\frac{\sin x}{x} = \prod_{k \geq 1} \left(1 - \frac{x^2}{k^2 \pi^2}\right)$.
2. Compare the coefficients of x^2 :
 - on the left, the Taylor series gives $-\frac{1}{6}$;
 - on the right, the product gives $-\sum_k \frac{1}{k^2 \pi^2}$.

Example. The same argument gives $\sum 1/n^4 = \pi^4/90$.

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Random points

Two ways to scatter points



exactly 40 points



$N \sim \text{Poisson}(40)$; here $N = 37$

Definition (Homogeneous Poisson process). For every region A , the number of points in A follows a Poisson law with parameter $\lambda |A|$, and disjoint regions receive independent counts [3].

Why Poisson?

Fine grid

Cut the window into m small cells; each one receives a point with probability $\lambda |A| / m$.

Taking the limit

The total count is binomial, and tends to $\text{Poisson}(\lambda |A|)$ as $m \rightarrow \infty$.

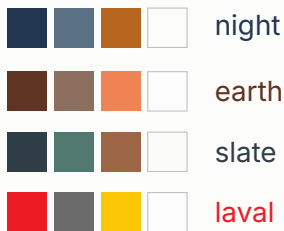
The starting point: **independent cells**. Everything else follows.

4

Toolbox

Four palettes

Chosen with a class option



- Primary, secondary, accent, background.
- Everything follows: titles, boxes, plots, links.

```
\documentclass[earth]{BeamerJS}
```



Cheat sheet

Environments

<code>defn</code>	definition
<code>thm, prop, lemma, coro</code>	results (optional name in brackets)
<code>proofbox</code>	proof
<code>ex, rem</code>	example, remark
<code>idea, notation</code>	icon boxes
<code>framedbox{Title}</code>	framed box with a free title
<code>keypoint</code>	the take-away sentence

Title page

<code>\position{...}</code>	job title
<code>\event{...}</code>	conference, seminar
<code>\titlelogo{file}</code>	logo on the left

Cheat sheet (continued)

Commands

<code>\term</code> , <code>\highlight</code> , <code>\stress</code>	emphasis
<code>\bigquote{quote}{author}</code>	displayed quotation
<code>\partialtoc[title]</code>	partial outline
<code>\NewBoxJS</code>	create a box
<code>\JSscatter</code> , <code>\JSpoissonScatter</code>	point clouds
<code>\palettepreview{name}</code>	palette preview
<code>\startappendix</code> , <code>\stopappendix</code>	appendix frames

Remark. Every name also has a French version (`\lemme`, `\lumiere`, `\defi`, `\planpartiel...`): see section 16 of `BeamerJS.cls`.

References

- [1] Martin Aigner and Günter M. Ziegler. *Proofs from THE BOOK*. 6th ed. Berlin: Springer, 2018.
- [2] Leonhard Euler. “De summis serierum reciprocarum.” In: *Commentarii academiae scientiarum Petropolitanae* 7 (1740), pp. 123–134.
- [3] John Frank Charles Kingman. *Poisson Processes*. Oxford: Oxford University Press, 1993.

Appendix: the tail of the series

Proposition. For every $N \geq 1$,

$$\frac{1}{N+1} < \sum_{n>N} \frac{1}{n^2} < \frac{1}{N}.$$

Proof. Use

$$\frac{1}{n(n+1)} < \frac{1}{n^2} < \frac{1}{n(n-1)}$$

and sum the two telescoping series. □

